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PROPOSAL FOR BOOKING APPOINTMENT SYSTEM

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For our project, we had to create a website with a booking appointment system for a clinic called "Klinik Ceria," near Batu Pahat, Johor. We are thankful to the clinic for allowing us to use their facilities and for Dr. Ummu, who kindly agreed to be interviewed for our research. Her insights into the workings of a clinic were truly invaluable, and we appreciate her generosity.

Our group consisted of five members who worked together to develop a functional and user-friendly website. We are grateful to each member of our team for their contribution and dedication to the project, which made it a success.

Finally, we would like to express our heartfelt appreciation to our lecturer, Dr. Cik Feresa Binti Mohd Foozy, for providing us with the opportunity to explore website development and design. Her experience and skills have been crucial to the success of our project. Her encouragement and support have been a source of motivation for us, and we are grateful for her guidance throughout our project journey.

In conclusion, we acknowledge the efforts of everyone involved in this project, including our lecturer, the clinic, and our team members. We are thankful for the knowledge and experience gained from this project.

Executive Summary

Our group assignment is about contacting outside organizations and create a website for the organization by studying the organization and negotiate access to the organization information. Therefore, we found a organisation which is a clinic called Klinik Ceria in Batu Pahat, Johor. Our group plan to create a website for them to make it easier for people to book appointments. Right now, people have to book appointments by calling or messaging the clinic on social media. This can be hard and takes a long time. Our group made a website called "Booking Appointment System" that helps people book appointments online.

We chose to use two software tools called WordPress and pantheon to make the website. WordPress is really easy to use and helps people make websites without needing to know a lot about coding. Pantheon is a tool that helps make WordPress even easier to use. It helps store information about the website and the people who use it. Pantheon is also free for people who are making a website for the first time.

Besides that, we talked to Dr. Ummu, who works at Klinik Ceria, to find out what the clinic needed in the website. We made sure to include everything that people and the clinic needed to make booking appointments easy. We also made sure the website was easy to use.

The website is also good for Klinik Ceria's staff. They can see all the appointments in one place, which makes it easier to manage. They can also see who is coming to the clinic and when. This means they can make sure the clinic is ready for people when they arrive.

Last but not least, the Booking Appointment System website makes it easier for people to book appointments with Klinik Ceria. It means people can book appointments from home, choose their doctor, and get reminders about their appointments. It also makes it easier for Klinik Ceria's staff to manage appointments. This project shows how technology can make healthcare better for everyone.

1.0 INTRODUCTION

1.0 Introduction of the Project

In today's hectic world, time is of the utmost importance, and individuals seek ways to optimize their daily routines. This is particularly true when it comes to medical services, where waiting periods can be long and irritating. Klinik Ceria, located in Batu Pahat, Johor, is a healthcare facility that provides quality care to its patients. The current appointment booking procedure at Klinik Ceria comprises patients scheduling appointments via phone calls and social media apps. This might result in extended wait times and miscommunication between patients and staff. Furthermore, this process can be time-consuming for both patients and staff, which can have a negative impact on the clinic's overall efficiency. The appointment booking system will be built with cutting-edge technology that is user-friendly and simple to use. It will come with a convenient interface that will enable patients to easily select their preferred appointment day and hour. Patients can arrange appointments at any time and from any location by deploying a digital booking appointment system. This will make booking appointments more convenient for patients while also freeing up staff time and enhancing the clinic's overall efficiency.

1.1 Problem Definition

Klinik Ceria at Batu Pahat, Johor is a healthcare facility that provides a health service to the patient. The clinic still uses the traditional method to book the walk-in appointment or contact the clinic's staff via WhatsApp to book the appointment. This traditional method can be time-consuming, inefficient, and miscommunication. The consequence is the long wait to see the doctor and the clinic's staff is struggling to manage the appointment efficiently. Besides that, the patient can easily cancel the appointment at the last minute. Other than that, it is very difficult for the patient to book an appointment in advance and choose the doctor for treatment. This project aims to develop an efficient system that meets the requirements of the clinic's staff and patients. This system also can give reminders for upcoming appointments to the patient.

1.2 Objective of the Project

The objective of this project is to solve the problems of Klinik Ceria. This company did not have enough professionals booking appointments system in daily. So, our group decided to create a website called “Booking Appointment System” that can help customers to book appointments more efficiently. One of the problems that Klinik Ceria has faced is that customers have to book manually which means they must book an appointment through social media or make phone calls. Since the customers are having difficulties in making appointments, the customers feel unhappy about it. So, we are creating the website based on the user’s needs. Besides, our group had an interview with Dr. Ummu and provided questions related to their company before developing the website.

At the end of this project, it also can increase the performance of the business of the company itself. In the meantime, we also used the waterfall model as our process model for this project. It is used as our sustainability guidelines. This helps us remain focused on the sustainability objective of our project. We hope that our website can fulfill the user’s needs and the user feels very happy and relieved when they use our website to make an appointment for their health.

1.3 Scope of the Project

The project is being undertaken by group 16 which consists of 5 members. The main aim of this project is to develop an online booking system that could help the patients to make an online booking and help the clinic manage the appointment. The scope of the project will contain a few phases such as planning, analysis, design, implementation, system security and support.

The planning phase is the first stage of the project. During the planning phase, all the group members will have a meeting to discuss the main goal for this project, and the timeline since the due date is already stated at week 14 and the scope of the project. After that, in this analysis phase, we also collaborate with the Klinik Ceria, Batu Pahat to determine the project requirement and resource allocation. The team members will identify the project resources to

deliver the project such as the equipment, hardware, and software. Moreover, all the team members are assigned the task of completing the project. Other than that, the leader of the group created the *Telegram* group chat to coordinate all the activities.

The analysis phase will involve the understanding of both Klinik staff and the patient for Klinik Ceria. The group also will perform a thorough review of Klinik Ceria's present appointment scheduling procedure. For the design phase, the group will design an online booking system based on the requirements from the Klinik Ceria. After the design phase is the implementation phase in which all the group members will be involved in developing the system using tools, software, and hardware. The final stage is system security to focus on security and ensure that the online booking system is secure and reliable.

The deliverable for this project is we will develop an online booking system that is easy for patients and staff to use for booking an online appointment and have the health service or treatment at Klinik Ceria. The system design also includes the database patient. The project teams will ensure the online booking system will be completed on time and meet the needs of Klinik Ceria staff and patients.

The assumption is to make sure that all the group members are successful in developing an online booking system for Klinik Ceria. Other than that, the online booking system will run smoothly without any errors and mistakes on the system. Another assumption is that the system will be subject to all relevant data privacy and security laws and rules, and necessary measures will be made to protect it against patient data leaks and other data breaches.

1.4 Significance of Project

The implementation of an appointment booking system for Klinik Ceria in Batu Pahat, Johor, will benefit the clinic and its patients. Patients will have better access to medical services as an outcome of a simpler and more effective booking process, leading to better health outcomes. The system will also help reduce patient wait times, bringing in increased patient

satisfaction and commitment to the clinic. Furthermore, the appointment booking system will improve the clinic's overall efficiency by reducing staff workload and improving patient appointment management. Staff may focus on providing quality care to patients by automating the appointment booking process, resulting in better health results.

The development of a digital appointment booking system is also in line with the trend of digital transformation in healthcare. The technology will offer a more modern and streamlined approach to appointment scheduling, which is becoming increasingly vital as more patients seek out healthcare services online. Overall, implementing a booking appointment system for Klinik Ceria will benefit both the clinic and its patients, improving the quality of care while increasing the clinic's efficiency and reputation.

2.0 LITERATURE REVIEW

2.1 Basic Concept, definition, and Theory

Online booking appointment systems are becoming increasingly popular in the healthcare business, offering patients a convenient and quick way to arrange appointments with healthcare providers or clinics. This literature review aims to provide an overview of the fundamental concepts, definitions, and theories related to online booking appointment systems for clinics. Online booking appointment systems are web-based platforms that allow patients to book appointments with healthcare providers online anywhere and anytime. These systems typically integrate with the provider's scheduling software, enabling patients to view available appointment times and book a slot that suits their schedule. Online booking appointment systems are designed to simplify the appointment booking process, easing the administrative burden on healthcare providers and improving patient satisfaction. Therefore, booking an appointment in a clinic becomes an excellent convenience for patients. It not only guarantees prompt medical attention but also enhances their overall satisfaction with the clinic. There are different ways to book an appointment in a clinic. For example, visiting the clinic in person is the most common way, such

as calling the clinic, or using online booking systems. Each method has advantages and disadvantages, and clinics should select a system that meets their objectives while also being simple for patients. It is vital to create and implement an effective booking system that benefits patients as well as the clinic. Regular reviews and improvements are also necessary to ensure the system remains effective over time.

2.2 COMPANY PROFILE

2.2.1 Klinik Ceria Background

Klinik Ceria has been operating for the past 2 years. It was established on 1st June 2021. It's a new clinic around Batu Pahat. This clinic operates from Monday to Saturday from 8:00 am until 10:00 pm and rests twice a day, from 1:00 pm until 2:00 pm and 7:00 pm until 8:00 pm. Sunday and the public holiday would be their day off. Moreover, they also served more than 2,000 patients monthly due to their warm services, reasonable price, and professionalism. Besides, they also registered as a vaccination center (PPV) for covid-19 vaccine. Klinik Ceria also provides products and services of healthcare for adults and children. Next, this clinic also has a good layout inside the clinic which includes registration and dispensary counters, a waiting area, a treatment room, 2D - 5D scanning, and a vaccination area. Klinik Ceria always prioritized the comfortability of its customers.

Most customers bring their family members and children. It came out with an open-shoes concept, a children's playground, a big space for a waiting area, free Wi-Fi, and an air-conditioned waiting area for our customers to feel happy and comfortable while waiting for treatment. We also provided 2 Android TVs and Coway Water Dispenser so that our customers can feel the environment like home. They are very genuine in providing healthcare services and solutions and try to give the best price that is affordable to the customers, especially for B40 and M40 household income. In Batu Pahat, Klinik Ceria is well-known for its affordable price and no hidden charges. They always will inform customers of the first prices before any treatment begins and will make sure patients know about the charge they need to pay before any procedures are done. This method is better for those who are struggling with financial problems.

2.2.2 History of Klinik Ceria

Klinik Ceria has been operating for 2 years and was established on 1st June 2021. They also served more than 2,000 patients monthly due to their warm services, reasonable price, and professionalism. Klinik Ceria is located in Batu Pahat, Johor. Most clinics in Batu Pahat operate during office hours. Thus, they provide solutions by being open from 9.00 am to 10.00 pm from Monday to Saturday so that people in Batu Pahat can get their appointment even at nighttime. They also provide house call services and any appointments if needed. Customers need to make phone calls or through social media to book an appointment. This method is old and very unused in this era of globalization. Klinik Ceria also provides almost all types of medicines for their patients. They will try their best to find customers at affordable prices for special medicines. This clinic really took care of the facilities and the quality of the product and provided good services to their customers.

2.2.3 Location and Environment

Klinik Ceria is located in Batu Pahat, Johor. It is beside the Old Street Commercial Centre building, which is the most famous and elegant building in Batu Pahat town. This clinic is located in the middle of the second row of the shop building with a big road and has a lot of parking lots. Due to its very strategic location, this clinic can easily access the customers.

2.2.4 Vision and Mission

A mission statement for this company is they always strive to provide the best and most friendly service, optimizing the convenience and ease of cheerful clinic action and maximizing the knowledge and quality of medical services of cheerful clinics while a vision statement for this company is “A top women and children-friendly clinic”.

2.2.5 Klinik Ceria Organization Structure

Klinik Ceria has two doctors, Dr. Ummu Asma' and Dr. Rozila. Dr. Ummu Asma' is the sole proprietor and person in charge while Dr. Rozila is the resident doctor of the Klinik Ceria. She is a young person, a warm-hearted, very talented doctor with multiskills, and very passionate about treating and caring for her patients. Her educational background, she graduated from Hasanuddin University, Makassar with first-class honors in 2016. Currently, she is attending NIOSH Course and Training to become an Occupational Health Doctor with her working experiences, in 2018 - 2019 as a house officer at Hospital Sultanah Nora Ismail, and 2019-2020 as a medical officer at Hospital Sultanah Nora Ismail.

Klinik Ceria had four young female staff with warm-hearted, friendly, high productivity, and willing to help. They are very open-minded with high integrity. They are also experienced, professional, and knowledgeable. They are young and passionate about learning new things and fast learners. As they want to give the best service, we only took the highly educated staff with a medical background as staff and paid them handsomely to give the best for their customers and They did great as expected. Besides, the customers also give a lot of good feedback on their services.



Figure 2.2.5 : Organization Chart

2.2.6 Klinik Ceria Products and Services

Klinik Ceria had almost all types of medication. For special medicine, they will try their best to find customers at affordable prices. They will order based on customer requests for their own benefits and health. This clinic also provides simple investigations (rapid tests) at affordable prices. Therefore, they will try to always give their best to serve our customers without prejudice. Klinik Ceria provides 26 facilities and 18 services for patients.



Figure 2.2.6(a) : Facilities of Klinik Ceria

SERVICES	CHARGES	SERVICES	CHARGES
ULTRASOUND 2D-5D SCANNING	RM30-RM45	CREAM,OITMENT,SHAMPOO	RM5-RM30
ECG	RM25	SUPPOSITORY/PESSARY	RM2-RM10
NEBULIZER	RM20	FEMALE CIRCUMCISION	RM40
SUCTION	RM20	MALE CIRCUMCISION (>4 YEARS OLD)	RM150-RM200
INTRAVENOUS DRIP	RM50	EAR PIERCING	RM30
INJECTION(MEDICINE AND VITAMINS)	RM10-RM30	SIMPLE BLOOD TEST AND URINE TEST	RM5-RM25
VITAMINS	RM0.30-RM2(PER PIECES)	BLOOD TEST (LAB)	RM20-RM350
MEDICINES(TABLETS AND CAPSULES)	RM0.30-RM10(PER PIECES)	MINOR SURGERY	RM50-RM200
SYRUPS	RM5-RM25	SWAB TESTS	RM50-RM170

Figure 2.2.6(b) : Services of Klinik Ceria

2.2.7 Current Project at Klinik Ceria

Klinik Ceria will produce new machines in their clinic to improve the facilities and services for the patients. The machine mentioned by Klinik Ceria is used to check the health of baby jaundice. It is a good investment in producing an advanced machine in the clinic. It also can attract more patients to come and check their baby's health at this clinic. Moreover, Klinik Ceria also organizes programs related to healthcare around Batu Pahat and the village area. This idea is a great idea because it can give awareness about the importance of maintaining their health. Besides, it also can increase and improve the name and performance of the clinic.

2.3 Existing System Investigation

A newly established clinic in Batu Pahat, Klinik Ceria provides healthcare services and solutions. They are well-known for their affordable price and no hidden charges which are one of the best choices of clinics for B40 and M40 household income. Klinik Ceria will always make sure to inform its patients about the prices and charges before any consultation, treatments, and procedures begin. They have provided about 18 services and 2-way consultation which is walk-in and house-visit.

As for making appointments, they use a call system to arrange patients' appointments. This system can cause problems such as overlooked calls because it is a human system that needs to pick up calls and schedule appointments and sometimes certain people will only call once and then they will proceed to another clinic which will cause the clinic to lose customers. Most of us do not want to communicate through calls as they have a packed schedule so they want to book an appointment online and just pick a date for their treatment. As for references, customers can only refer to their social media such as Facebook, Instagram, and TikTok which can be hard to access to anyone who does not have any social media and it is sometimes kind of hard to find specific information as we need to scroll one-by-one.

Klinik Ceria only has one doctor in charge and one resident doctor so it will be difficult and takes time to arrange appointments. Especially services such as blood tests, urine tests, and swab tests which are not emergency cases that can be scheduled in advance for an appointment. It will be easier for doctors to arrange their schedule and then customers can just choose when they are free to do their treatments. This problem can be solved by creating a system and a website which is also a great way to keep a database of their customers.

3.0 SYSTEM DEVELOPMENT METHODOLOGY

3.1 Introduction

System development methodology is a process that refers to planning, analysis, design, implementation, and system security and support. The system development methodology provides a structured approach to building a high-quality system that can meet the requirements of the Klinik Ceria, Batu Pahat. To develop the online booking system, it is required to have a systematic process to achieve a good, secure, and user-friendly system for the Klinik Ceria. There are a few examples of methodologies including "agile software development," "water-fall development," and "spiral development." Waterfall development is a sequential development method that moves through all project phases such as planning, analysis, design, implementation, and system security and support like a waterfall, with each phase ending up entirely before the next one starts, and it is a traditional method and suitable for developing an online booking system. Spiral development is an approach to risk management that combines aspects of the Waterfall model and the iterative method of the development model. Agile software development is to improve client satisfaction, fast release small software that works parts. The methodology must involve all the requirements of staff in Klinik Ceria and the patient that will use the system.

It's important to note that project methodologies for website implementation can vary depending on the organization's preferences and the complexity of the project. Some common methodologies used in website development include Waterfall, Agile, or a combination of both (hybrid approach). In conclusion, implementing a project methodology in system analysis and design for implementing a website ensures a structured and systematic approach to website development. It guides the team through requirements gathering, system design, development, testing, and deployment phases. By following a well-defined methodology, organizations can increase the chances of successfully delivering a website that meets user needs, aligns with business objectives, and provides an optimal user experience.

3.2 Software Process model adopted in the project

The waterfall model is an approach to the software development lifecycle (SDLC) that is linear and sequential. The waterfall model follows a logical order of SDLC stages for a project, much to the way water flows over a cliff's edge. It establishes specific results or objectives for every stage of development. After being achieved, certain endpoints or objectives cannot be changed.

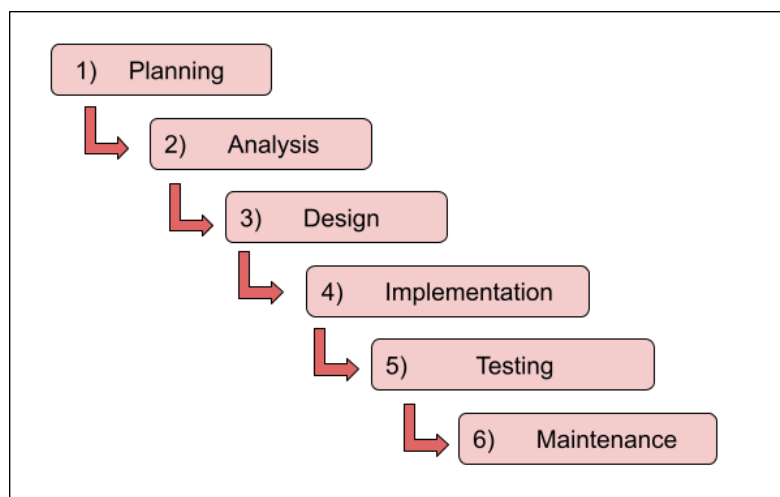


Figure 3.1: Waterfall Model

The Waterfall model is one of the models in the System Development Life Cycle. The advantage of this model is it can identify the requirements long before programming begins and limit changes to the requirements as the project proceeds. It is an approach in which the entire process of software development is divided into separate phases. This model moves from one phase to another and moves forward in the same manner as the waterfall.

For the planning phases and analysis phases, the requirements of this system to be developed are stored in this phase and documented in a requirement specification document. Besides, our group also had an interview session with the clinic for the analysis phase. Next, in the designing phase, the requirement specifications from the first phase are studied in this phase and the system design is based on the user's needs. This system design also helps in determining hardware and system requirements and assisting in defining the overall system architecture. Moreover, in the implementation phase, we begin the project by developing the system using WordPress. Each button and requirement in the system is developed and tested for its functionality, referred to as the testing phase. Finally, after developing the system, we must maintain and update the system to fix the issues, and patches are released. Also, to enhance the product some better versions have been released. Maintenance is performed to deliver these changes in the customer's environment.

3.3 Project Planning

The first steps in developing a new system are to have proper project planning. In this phase, we will list all tasks that need to be done and assigned to each team member. By using Software Development Life Cycle methodology (SDLC), we need to estimate the time required to complete each task. Also, we need to schedule the project wisely so that we get to complete the project using the timeline we have planned. After our group discussion, we made a Gantt Chart to help us visualize the timeline of the project.

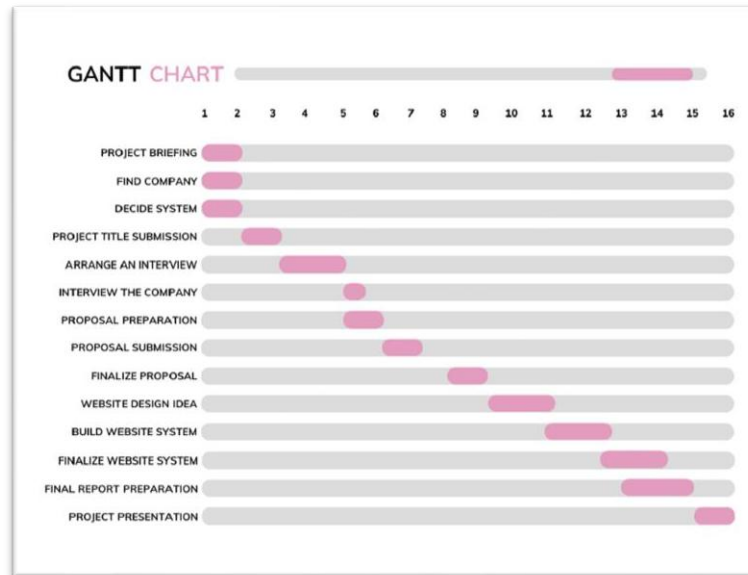


Figure 3.2 : Gantt Chart

3.3.1 Feasibility Study

A feasibility study is an organised analysis of the viability and potential for success of a proposed project, system, or endeavour. A feasibility study's goal is to give decision-makers with important insights and information that will allow them to make educated decisions about whether to proceed with the project or seek alternative possibilities.

A feasibility study is an in-depth study of a proposed project or system to assess its viability and probability of success. Here are some crucial variables to examine in a feasibility study for Klinik Ceria's booking appointment system. A feasibility study for the introduction of a booking appointment system for Klinik Ceria, a healthcare facility, including a thorough and methodical examination of the project's viability and potential for success. Technical feasibility, economic feasibility, and operational feasibility are three important elements to evaluate in this study.

3.3.1.1 Technical feasibility

The feasibility study's technical feasibility assesses the practicality and viability of implementing the proposed system. In the instance of Klinik CERIA's appointment booking system, the system is being built with WordPress, a popular content management system (CMS). WordPress provides a stable and adaptable framework for developing websites and applications, making it ideal for developing the appointment booking system. The broad selection of themes, plugins, and customization possibilities available with WordPress can improve the development process, allowing the establishment of a user-friendly and visually beautiful system.

The WordPress CMS makes it simple to handle material such as appointment scheduling, patient information, and other pertinent data. Its user-friendly interface and thorough documentation make it accessible to developers, allowing for more efficient system development and maintenance. Furthermore, WordPress is a highly scalable platform that can accommodate Klinik CERIA's development and rising needs over time.

In terms of hardware, the system may be accessible via computers and cellphones. Because each team member has their own laptop, there is no need to borrow or rent extra machines, streamlining the development process. The website is adaptable, adjusting to multiple screen sizes and orientations to provide a consistent user experience on both computers and smartphones. Users may simply access and operate the system without device constraints by delivering a uniform design and functionality across devices.

The technical feasibility of Klinik CERIA's booking appointment system is ensured by using WordPress as the selected platform and taking into account the hardware requirements for multi-device accessibility. The use of WordPress's extensive CMS features, scalability, and reactivity guarantees the building of a user-friendly and flexible system that fits Klinik CERIA's and its patients' demands.

3.3.1.2 Economic feasibility

Several criteria must be examined when analysing the economic feasibility of embracing the booking appointment system for Klinik Ceria in order to establish its financial sustainability and justify the expenditure. It is essential to examine both tangible and intangible benefits and costs associated with the project

Tangible benefits of our booking website include the increased efficiency and convenience it brings to both patients and the clinic. With the website, patients can easily schedule appointments online, saving time and effort compared to traditional methods. This streamlined process enhances patient satisfaction and improves overall accessibility to healthcare services. Additionally, the website allows the clinic to effectively manage appointments, reducing administrative tasks and optimizing resource allocation.

On the other hand, because we chose a free version of WordPress, the actual costs related to the website's installation are substantially lower. As a result, there is no longer a requirement for substantial financial investments in site design, development, hosting, and ongoing maintenance. We can efficiently administer and maintain the website at zero cost by utilizing the tools and resources provided by the free edition of WordPress.

The booking website brings intangible benefits in the form of an enhanced patient experience and an improved reputation for Klinik Ceria. Through its user-friendly interface and seamless appointment booking process, the clinic is perceived as modern, efficient, and dedicated to patient care.

However, when the booking website is being implemented, intangible costs start to appear. The costs involved, which include things like the learning curves and adaption process for clinic staff, are difficult to quantify. To acquaint employees with the new system and guarantee a seamless integration into their current workflow, training sessions may be required. To address any problems or worries that develop during the transition phase, ongoing staff help and monitoring may also be necessary

3.3.1.3 Operational Feasibility

Our website proposal for Klinik Ceria must be operationally feasible in order to be implemented successfully and sustainably over time. We thoroughly assessed a number of variables to determine whether the website could be operated.

The accessibility of the resources required to operate the website is a crucial factor. In order to manage the anticipated traffic and data transactions, this entails having a dependable hosting provider, sufficient server capacity, and strong internet access. We can ensure seamless website performance, reduce downtime, and improve user experience overall by putting these resources in place.

The website's compatibility with Klinik Ceria's current systems and infrastructure is another factor to take into account. The website's interaction with the clinic's appointment scheduling software, patient database, and other appropriate systems has been considered. This seamless integration eliminates human information duplication and lowers the possibility of mistakes while enabling effective data interchange and synchronisation.

The operational effect on the clinic staff has also been evaluated. We have equipped the employees for the launch of the website and its related procedures through training and assistance. We have made sure they have the knowledge and abilities needed to use the system, schedule appointments, and handle any potential technical difficulties. We can reduce disruptions and guarantee an efficient process for the clinic by supporting a smooth changeover.

Finally, the website's usability and straightforward design contribute to its operational viability for patients. The user interface has been considered, and we've made sure it's easy to use, understand, and visually appealing. Patients may quickly explore the website, locate the information they want, and make appointments without any difficulty because to the user-friendly interface.

3.3.2 Data Gathering

Data Gathering is essential to the growth of a project since it forms the basis for strategic planning and well-informed decision-making. It entails using a methodical strategy to gather, prepare, and analyse data in order to draw forth important knowledge and insights. The main goal of data collection is to acquire reliable and pertinent information that will provide a thorough knowledge of the problem or requirements at hand. Using this data as a starting point, we may define the project's scope, determine its goals, and create efficient methods for overcoming obstacles and achieving the objectives.

3.3.2.1 Method Carried Out for Data Gathering

It is important to select the appropriate data-gathering methods based on the research objectives, available resources, and the nature of the data being sought. Often, a combination of methods is used to gather comprehensive and diverse data to address research questions or objectives effectively. Based on what we have learned during class, there are some elicitation techniques that we can choose those methods for data gathering before implementing the website. There are methods in the elicitation techniques such as interviews, document analysis, observations, questionnaires, and JAD sessions.

So, for our project, we chose to use the interview and document analysis methods. For interviews, we choose to conduct interviews with Dr. Ummu so that she can provide valuable insights into the needs, requirements, and expectations. Structured or semi-structured interviews can also be used to gather information about current processes, pain points, and desired improvements. Interviews can be structured with a predetermined set of questions or unstructured more conversational and flexible. They provide an opportunity to gather in-depth information from individuals or groups and allow for follow-up questions to explore specific topics further. In the interview sessions, our group also have done surveys and questionnaires. We already prepared the questions based on the website that we want to develop before the interview.

Next, the other method that we applied in the project is document analysis. This method involves reviewing and analyzing existing documents, such as reports, articles, records, or archival data. It can provide valuable historical or contextual information. After the interview sessions with Dr. Ummu, our group reviewed the data recorded through the interview journey. We extract information during the document analysis, key information is extracted from the documents. This can include data entities, attributes, relationships, process flows, user roles, system constraints, business rules, and any other relevant information. The extracted information helps in understanding the current system and identifying areas for improvement. Overall, document analysis is a valuable method in system analysis and design as it provides a comprehensive understanding of the existing system or processes for our project. It helps in gathering relevant information, identifying gaps and inconsistencies, and providing a foundation for designing an improved and efficient system.

3.3.2.2 Analysis of Collected Information

3.3.2.2.1 Analysis of Interview

In this project, we are using the elicitation technique of interview to collect data from the clinic named 'Klinik Ceria 'and ask for their permission to create a website for them. The interview reveals the results of a thorough conversation with Dr. Ummu of Klinik Ceria that was done to obtain important data for the creation of the clinic's new website. A system for scheduling appointments will be the focus of the website, which is intended to make things easier for both the clinic and its clients. The purpose of the interview was to learn about the clinic's preferences and needs for the website, including its functionality, design, and content.

In the interview, the clinic expressed a preference for a multilingual website, emphasizing Bahasa Melayu and Mandarin as the primary languages. This requirement is crucial to ensure the website is accessible and user-friendly for the patients who live around the area, Batu Pahat where there are more Malay and Chinese citizens. We also know that Klinik Ceria is staffed by two doctors, with one handling the weekday appointments and the other covering nights and

weekends. This information is vital for the appointment booking system, as it will allow patients to book appointments based on the doctor's availability.

At Klinik Ceria, they don't specialize in any specific medical field, but they focus on providing excellent family doctor services. Patients will have the option to choose their preferred doctor when they visit the website, which is great! The clinic operates on a schedule that allows for as many slots as possible, and each patient's appointment typically lasts for 30 minutes. They're closed on Sundays, so plan accordingly. Klinik Ceria is dedicated to sourcing special medications for their customers at affordable prices, which shows that they really care about their patients' well-being. This commitment to patient care will be highlighted on the website so that everyone knows how much they value their patients.

The clinic wants the website to be simple and easy to use, not overcrowded with information. This is great news for patients who just want to book an appointment quickly and easily. The website will be accessible on a variety of platforms, including Facebook, Google, WhatsApp, TikTok, and Instagram, which is convenient.

Klinik Ceria is committed to keeping patients' personal information safe and secure. They will follow data protection regulations to store patients' names, IC numbers, gender, and phone numbers. The website will have a calendar to make booking appointments convenient. The website's colors will be light pink or pink with white, matching the clinic's branding. Patients will be able to see the doctor's name on the day of their appointment, which is transparent and helpful.

The clinic values patient feedback and will display their email address prominently on the website. This feedback system will help the clinic improve its services based on patient suggestions. Patients will need to provide their name, IC number, gender, and phone number when making an appointment. This information will be collected securely through the appointment booking system.

The website design will be user-friendly, accessible, and functional. Klinik Ceria

prioritizes its patients' experience, whether they visit the clinic or use the website.



Figure 3.3.2.2.1(a) : Before the Interview Session



Figure 3.3.2.2.1(b): After the Interview Session

3.3.2.2.2 Analysis of Document Analysis

In our project, we have the honor to get an official document from the 'Klinik Ceria'. Therefore, our group is using another elicitation technique, document analysis, to gather our information to create a website suitable for the clinic's needs. The document provided about Klinik Ceria is structured in a clear and organized manner, beginning with a brief introduction about the clinic, its location, and its mission and vision statements. It then delves into the qualifications and experience of the doctors and staff, followed by the services and facilities offered by the clinic.

The information is thorough and covers the clinic's location, the training, and experience of the doctors and staff, the facilities and services provided, and the clinic's dedication to providing reasonably priced healthcare. The clinic's mission and vision are also highlighted in the document, which demonstrates its dedication to offering top-notch, patient-centered treatment.

The document's apparent goal is to provide information about Klinik Ceria to potential clients or employees. It offers thorough details that are helpful for anyone attempting to understand the clinic's offerings, staff, and overall approach to healthcare. The paper appears trustworthy in terms of veracity because it contains detailed information on the doctors, including their names, educational backgrounds, and professional experiences.

The document seems to favor Klinik Ceria in a favorable way. It frequently promotes the clinic's advantages, including its advantageous location, skilled and knowledgeable personnel, and broad range of services. Although this is typical in promotional materials, it's significant to note that neither difficulties nor opportunities for improvement are mentioned in the document.

Overall, the document suggests that Klinik Ceria is a well-established, patient-centered clinic that provides high-quality care.

3.3.3 Gantt Chart



Figure 3.5: Gantt Chart

3.4 Software and Hardware Requirements

List of hardware	Specification	Purpose
Smartphone devices (any moveable devices)	1. Able to connect to the internet. 2. Meet all the required specifications for the system.	1. To run the program. 2. Gather data from team members
Laptop	1. Display the number of doctors 2. Display the available timetable for each doctor	1. To create a system. 2. To test and run the system. 3. Fix the system if any errors occur.

Table 3.4 (a). List of hardware

List of software	Specification	Purpose
WordPress	1. Create and manage a website without needing to have extensive coding knowledge	1. To create a website 2. To design a website

● Pantheon	1. Store and manage the website's data 2. Store user information, appointment details, and other relevant data.	1. To create a Database for the system.
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Table 3.4 (b). list of software

When developing a system in system analysis and design for this project, it is crucial to determine the software and hardware requirements. The software and hardware requirements provide the foundation for the system's development and deployment. Here are some considerations for each list of hardware such as smartphone devices (any moveable devices). For smartphone devices, we use it as our platform to gather the data or input for any team members. Moreover, we also use smartphones to contact the clinic for any work that is related to this project. However, by using smartphones we are able to connect with the internet since most of our projects are online. Other than that, we also can meet all the required specifications for the system.

Moreover, the list of hardware that we use for the project is laptops. We use this hardware most of the time compared to smartphones. This is because the laptop can create the system by using software that we have been chosen for our project. Then, it also functions to test and run the system and fix the system if any errors occur. Laptops have better and larger space to develop the system than smartphones. So, we use laptop to create the website and through this hardware we can display the number of doctors and display the available timetable for each doctor.

Next, we moved to the list of software we used to implement the website. We used WordPress to develop the system. The reason why we chose WordPress as a platform to create the website is the functionality of the software is easy to use and it is user-friendly software. We can edit the website in a time together and the most important part is, it is free to use it. WordPress also can create and manage a website without needing to have extensive coding knowledge. Lastly, we use pantheon software. It functions as a plug in to WordPress and make more easier to edit and choose the best template for our website. The specialty of it is it can store and manage website's data and store the user information details such as, appointment details and other relevant data. Pantheon also serves free databases for the first-time users in developing the website.

It is essential to conduct a thorough analysis of the system requirements, consult with relevant stakeholders, and consider scalability, security, and performance factors when defining the software and hardware requirements for the system. This helps ensure that the system is designed and implemented with the necessary software and hardware components to meet its objectives effectively.

4.0 ANALYSIS & DESIGN

4.1 Introduction

The analysis phase of the System Development Life Cycle (SDLC) is an essential one where requirements are collected, system objectives are specified, and present problems have been analyzed. Additionally, business needs are gathered during the analysis phase, along with any important risks. A feasibility study is also part of this SDLC stage that determines all project strengths and weaknesses to evaluate the project's overall viability. Entity Relationship Diagrams (ERD), Data flow diagrams (DFD) and process specification forms will be used to analyze the data during the analysis phase.

Two frequently used methods for business systems analysis are entity-relationship diagrams (ERDs) and data flow diagrams (DFDs). It will be helpful to view, record, and represent a system's interactions, data, and operations. DFDs illustrate how data flows through the system, how it is transformed, and how it communicates with other entities. ERDs represent the organization of data into entities, characteristics, and relationships and the limitations and regulations that control the data. Both diagrams use consistent and straightforward use of symbols, labels, and standards to represent information.

Next, the design phase is still in the System Development Life Cycle (SDLC). The technical requirements of the product are set out by software developers during the design process. Screen designs, databases, designs, system interfaces, and prototypes are a few examples of these details, depending on the project. These specifics help clients decide on the final product design. In this design phase, we also focus on Determining the system acquisition strategy. Next, the detailed design phase focuses on the way data will be stored, interface design input design and output design, create the system specification. This ensures developers understand how each component will be implemented.

However, it is important to focus on security measures, such as login functionality and reCAPTCHA, throughout the design phase of developing an online booking system. Ensure a secure method of authentication for the login functionality using methods like password and protection against common vulnerabilities. Use a reliable reCAPTCHA system to identify users as humans and stop automated attacks. Before development, prototyping can help in verifying these security measures and collecting input for improvement.

4.2 Analysis

The analysis phase is one of the most important parts of the project as it will help us to gain feedback and support from internal and external supervisors. In this phase, it will define the end-user requirements using the high-level requirements.

4.2.1 Data Flow Diagram and Process Specifications Form

4.2.1.1 Context Diagram

The definition of a context diagram is a graphic representation of a system that demonstrates how the system interacts with its surroundings. The scope of the system is defined, and the entities that interact with it are identified. In our project we are doing a booking appointment system for a clinic, therefore our group has drawn the context diagram which is able to show the four entities that interact with the system which are Doctors, Manager, Staff Nurses, and Patients in the figure below.

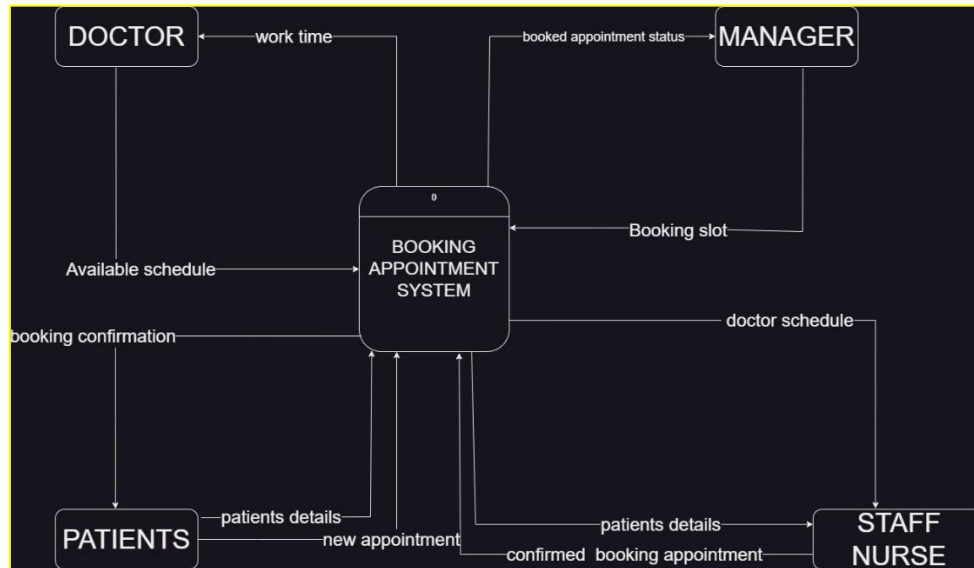


Figure 4.2.1.1(a): Context Diagram of Booking Appointment System

The first entity in the context diagram is doctors. Doctors are the primary users of the appointment booking system and rely on it to manage their schedules and appointments. The system allows doctors to view their appointments for the day, week, or month, and make changes to their schedules, as necessary. Additionally, the system may provide doctors with alerts or reminders for upcoming appointments to ensure that they are prepared and on time.

The second entity in the context diagram is the manager. The manager is in charge of overseeing the scheduling system and making sure everything runs well. The booking window for the entire clinic is the manager's primary responsibility. For instance, the manager has the authority to alter the clinic's operating hours, days of operation, rest days, and holidays. The management has access to reports on the system's performance, can keep an eye on how the system is being used, and can modify the configuration of the system as necessary. The management can also utilize the system to create reports or dashboards about how the system is being used and performing, which can be used to guide decisions and spot areas for development.

The third entity in the context diagram is the patients. Patients use the booking appointment system to schedule appointments with doctors. They can view available appointment times, and services and schedule an appointment. Patients can also use the system to

update their personal information, such as their address or phone number. After that, when the patients successfully booked the appointment, there will be a booking confirmation sent to each patient that has completed the booking process.

The fourth entity in the context diagram is the staff nurse. The staff nurse is responsible for approving the schedule that the patients' appointments and ensuring that patients are seen on time. The staff nurse can view the clinic's schedule, make changes to appointments as needed, and communicate with doctors and patients.

4.2.1.2 Level 0 of the Data Flow Diagram (DFD)

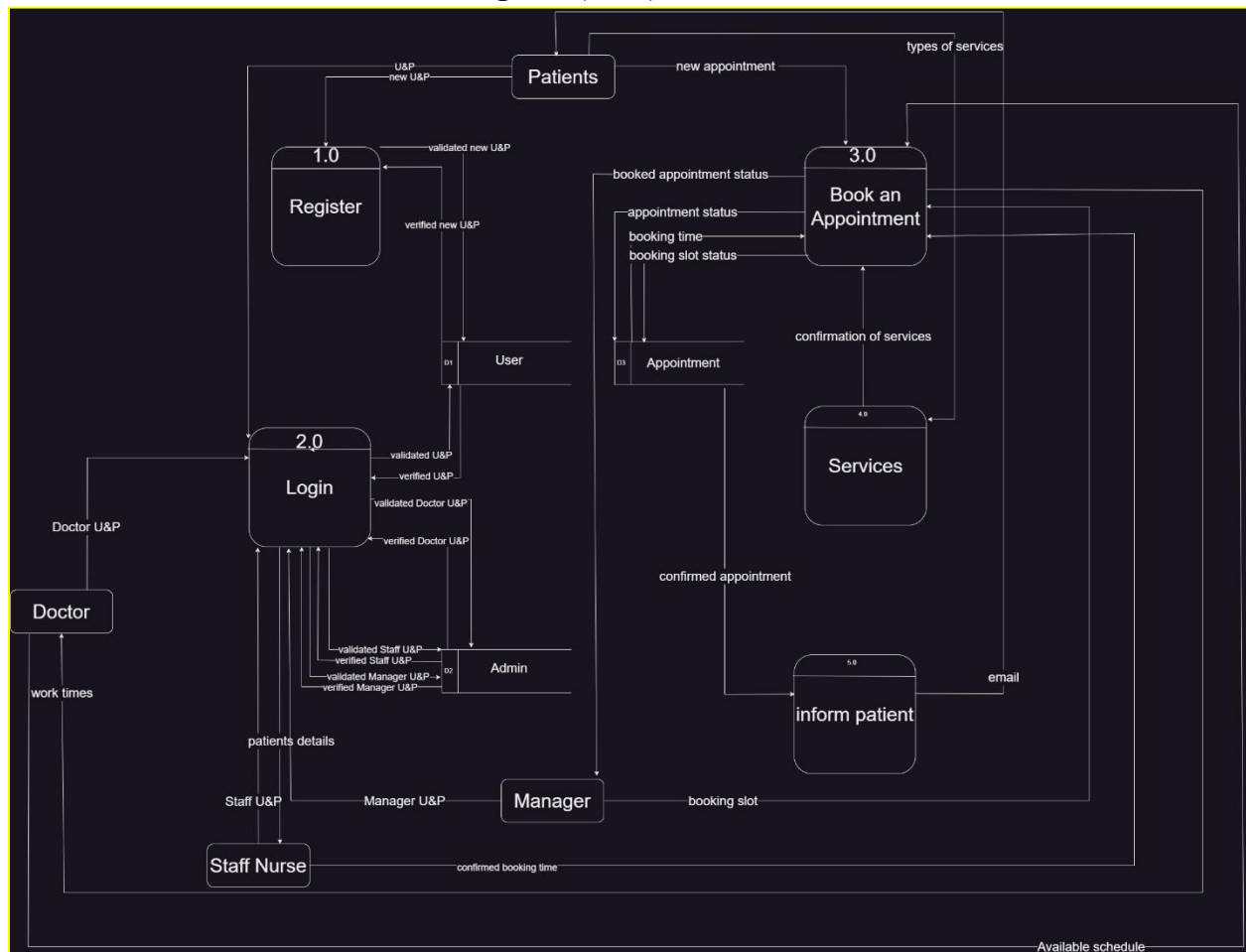


Figure 4.2.1.1(b): Level 0 of Booking Appointment System

In our project, we are doing a system called the booking appointment system for a clinic. The Data Flow Diagram (DFD) Level 0 shown above represents the interaction between four entities which are the Manager, Patients, Staff Nurse, and Doctor. These entities interact with the system through five key processes which are Register, Login, Booking an Appointment, Services, and Inform Patient. The system maintains three data stores that involve in the booking appointment system are Admin, User, and Appointment.

For patients who are new to the system or the first user for the website, they must first go through the Register process. This process involves the validation and the verification of their username and password, which are subsequently stored in the User data store. Patients who have already registered can directly access the system through the Login process using their existing username and password. For the Admin entities which include the Staff Nurse, Doctor, and Manager, also interact with the system through the Login process. Their personal details are stored in the Admin data store.

Once logged in, patients can proceed to book an appointment as the booking pages only access for the logged in users. The status of this appointment after booking is stored in the Appointment data store. The Staff Nurse plays a important role in confirming the booking time through the Booking an Appointment process. Once confirmed, the appointment details are stored in the Appointment data store and the Inform Patient process is initiated. This process involves sending an email to the patient to inform them of their confirmed appointment.

Patients also have another option to book an appointment, they can view the clinic's services before booking. This is facilitated through the Services process. After they selected the services, it will proceed back to book an appointment. Next, the Doctor, another key entity, provides their available schedule to the system through the Book an Appointment process and can view their work times. Lastly, the Manager, who oversees the overall operation, manages the booking slots for the clinic through the Booking an Appointment process. This ensures the smooth running of the clinic and efficient allocation of resources.

4.2.1.3 Level 1 Data Flow Diagram (DFD)

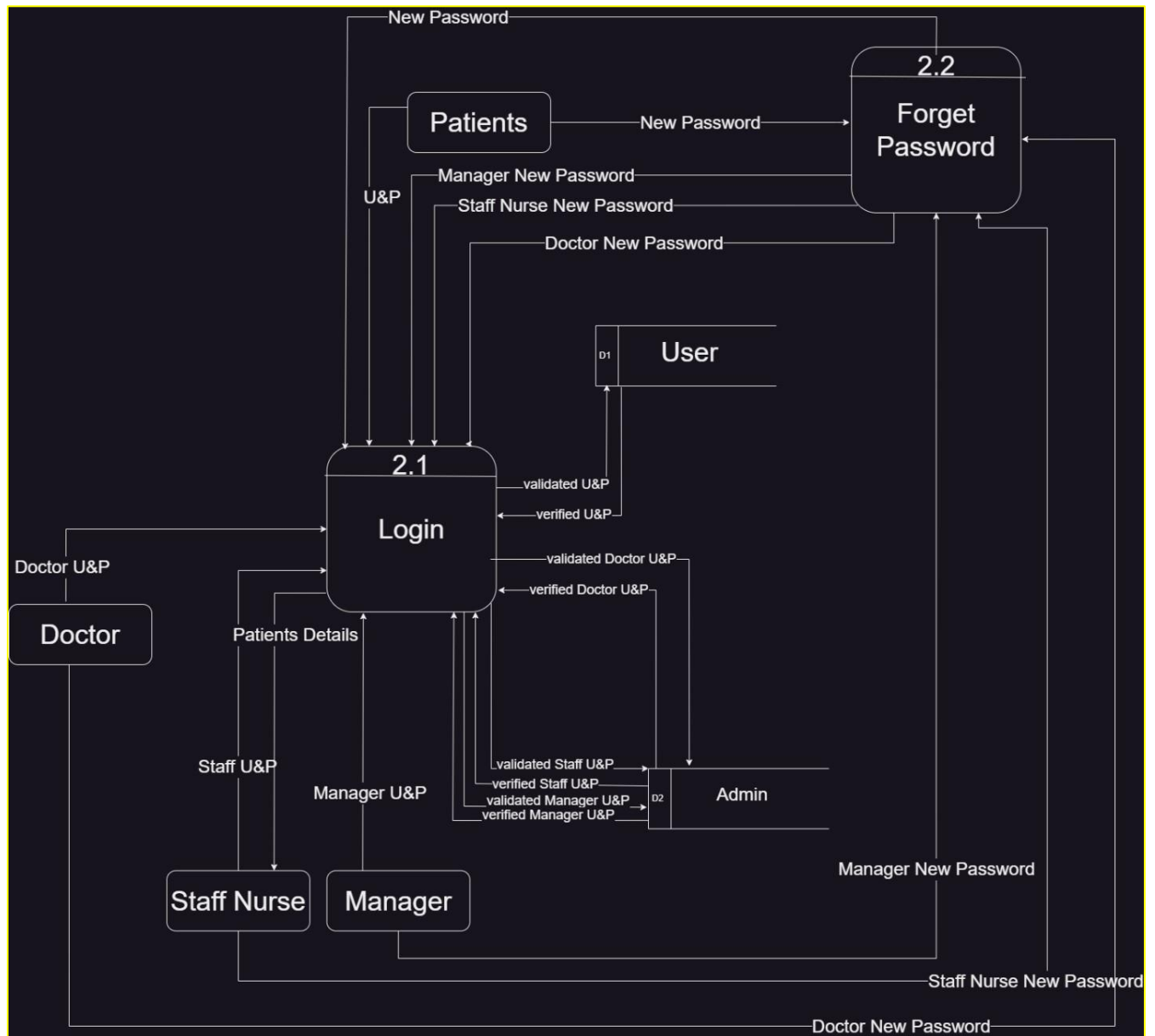


Figure 4.2.1.1(b): Level 1 of Login Process

In the Level 1 Data Flow Diagram (DFD), the processes are broken down into more detailed sub-processes. In this case, the “Login” process now includes an additional sub-process which is the “Forget Password”.

When a patient, doctor, manager, or staff nurse forgets their password, they can initiate the “Forget Password” process. This process is typically designed to help users recover or reset their passwords securely. It might involve the user confirming their identity through their email to reset their password. The user (Patient, Doctor, Manager, or Staff Nurse) initiates the “Login” process but cannot remember their password. They can select the “Forgot Password” option, which triggers the “Forget Password” sub-process. The system may ask the user to provide their registered email address. This information is used to identify the user's account in the User or Admin data store. The forget password process then sends a password reset link to the user's registered email address. The user uses this link or code to create a new password. The system validates the new password and updates the User or Admin data store with the new password. The user can now log in using their username and new password.

This additional sub-process ensures that users can regain access to their accounts even if they forget their passwords, enhancing the system's usability and security.

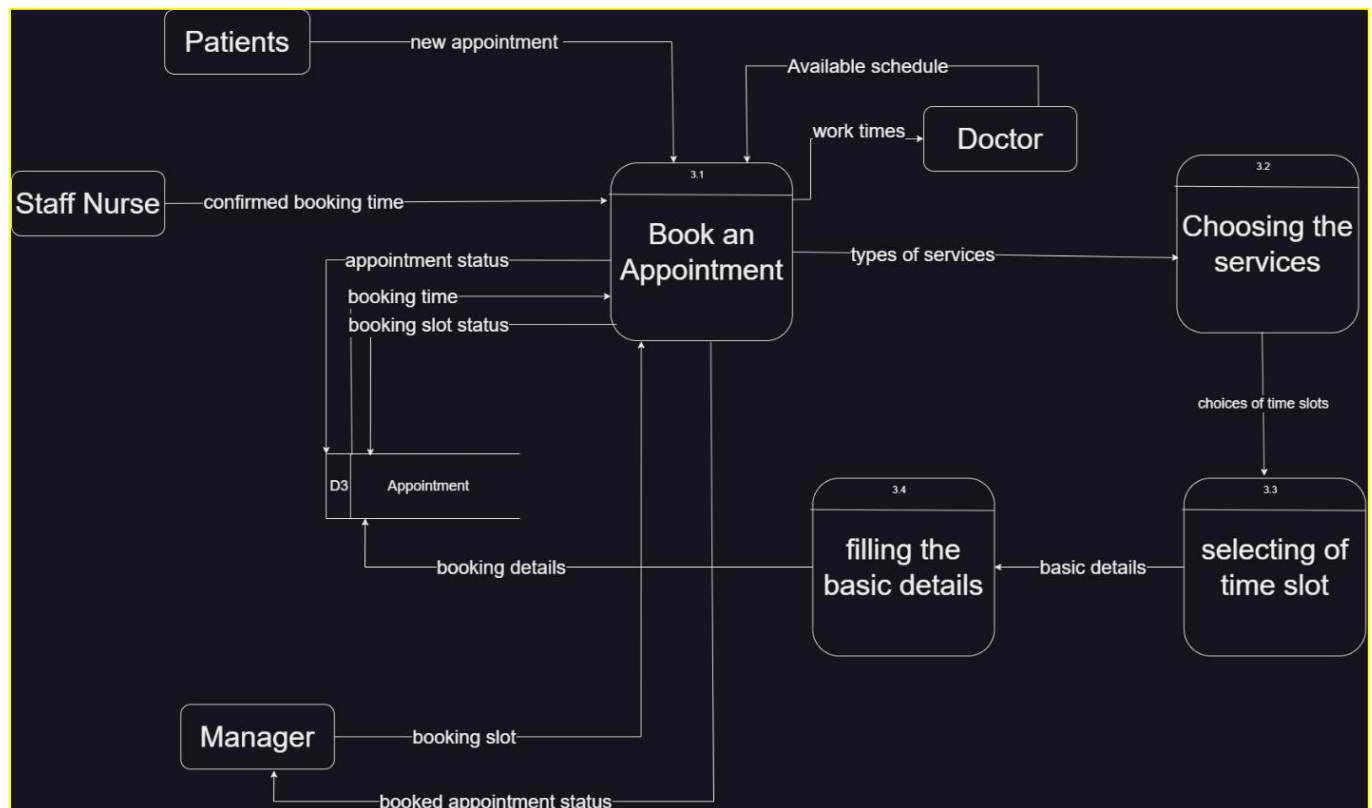


Figure 4.2.1.1(c): Level 1 of Booking an Appointment Process

The level 1 DFD of the book an appointment process expands and breaks it down into three sub-processes: "choosing the services," "selecting time slots," and "filling basic details." When a patient initiates the "book an appointment" process, it will show the first sub-process is "choosing the services." This means that the patient will be presented with a list of available services, and they will select the service they require. After the patient has chosen their service, then it will jump to the next sub-process is "selecting time slots." This process will be presented with a list of available time slots for the selected service to patients, and they will choose the time slot that works best for them. Once the patient has selected their time slot, it shows that the final sub-process is "filling basic details." Then the patients must provide their personal details, such as name and contact information.

Lastly, all booking details will be stored in the appointment data store. This is a data store that will keep track of all the appointments that have been booked.

4.2.1.4 Process Specifications Form for the Booking Appointment System

Process Specification Form	
Number: 1.0 Name: Register Description: For the user or the patients to sign up in order to make an appointment from the website.	
Input Data Flow: Username First Name Last Name E-mail Address Password Confirm Password Google ReCAPTCHA	
Output Data Flow: Approval of user or patient	
Type of Process: Online ☒ Batch Manual	Subprogram/Function Name:
Process Logic: IF The user or patient has no account THEN A member account is successfully registered ELSE Registration unsuccessfully END IF	
Refer to: Structured English ☒ Decision Table Decision Tree	
Unresolved Issues:	

Table 4.2.1.4(a): Register Process

Process Specification Form	
Number: 2.0 Name: Login Description: For the user or the patients who already have an account or user have successfully registered, they log in for booking an appointment	
Input Data Flow: Username Password Google ReCAPTCHA	
Output Data Flow: Redirect to the Homepage of the Klinik CERIA	
Type of Process: Online ☒ Batch Manual	Subprogram/Function Name:
Process Logic: IF The user or patient has an account THEN Login successfully ELSE Login unsuccessfully END IF	
Refer to: Structured English ☒ Decision Table Decision Tree	
Unresolved Issues:	

Table 4.2.1.4(b): Login Process

Process Specification Form	
Number: 3.0 Name: Book an Appointment Description: For the user or patients who already log in, can book an appointment with available time slots, services, dates and so on.	
Input Data Flow: Book an Appointment Types of services Choices of time slot Choices of date Basic details of patients	
Output Data Flow: Summary of Appointment Successful Page	
Type of Process: Online ☒ Batch Manual	Subprogram/Function Name:
Process Logic: IF The user or patient has an account THEN Able to book an appointment THEN Choose the services Pick the time slots Choose the dates Fill up the basic details THEN Successfully booked ELSE The booking page restricted END IF	
Refer to: Structured English ☒ Decision Table Decision Tree	
Unresolved Issues:	

Table 4.2.1.4(c): Book An Appointment Proces

Process Specification Form	
Number: 4.0 Name: Services Description: The user or patient after login, they are able view the services before booking an appointment	
Input Data Flow: Services Book an Appointment Choices of time slots Choices of date Basic details of patients	
Output Data Flow: Summary of Appointment Successful Page	
Type of Process: Online ☒ Batch Manual	Subprogram/Function Name:
Process Logic: IF The user or patient has an account THEN Able to view the services THEN Able to book an appointment THEN Pick the time slots Choose the dates Fill up the basic details THEN Successfully booked ELSE The booking page restricted END IF	
Refer to: Structured English ☒ Decision Table Decision Tree	
Unresolved Issues:	

Table 4.2.1.4(d): Services Proces

Process Specification Form Number: 5.0 Name: Inform Patients Description: When the patients successfully booked an appointment and are confirmed by the staff nurse of the 'Klinik CERIA', an email notification will be sent to inform patients	
Input Data Flow: Submit the appointment	
Output Data Flow: Confirmation of appointment Email Notification	
Type of Process: Online ☒ Batch Manual	Subprogram/Function Name:
Process Logic: IF The user successfully booked the appointment THEN The booked appointment will confirm by the staff nurse THEN The email notification will send to the patients through email ELSE Booking failed END IF	
Refer to: Structured English ☒ Decision Table Decision Tree	
Unresolved Issues:	

Table 4.2.1.4(e): Inform Patients Process

4.3 Design

4.3.1 Database Design (ERD & Related Notation)

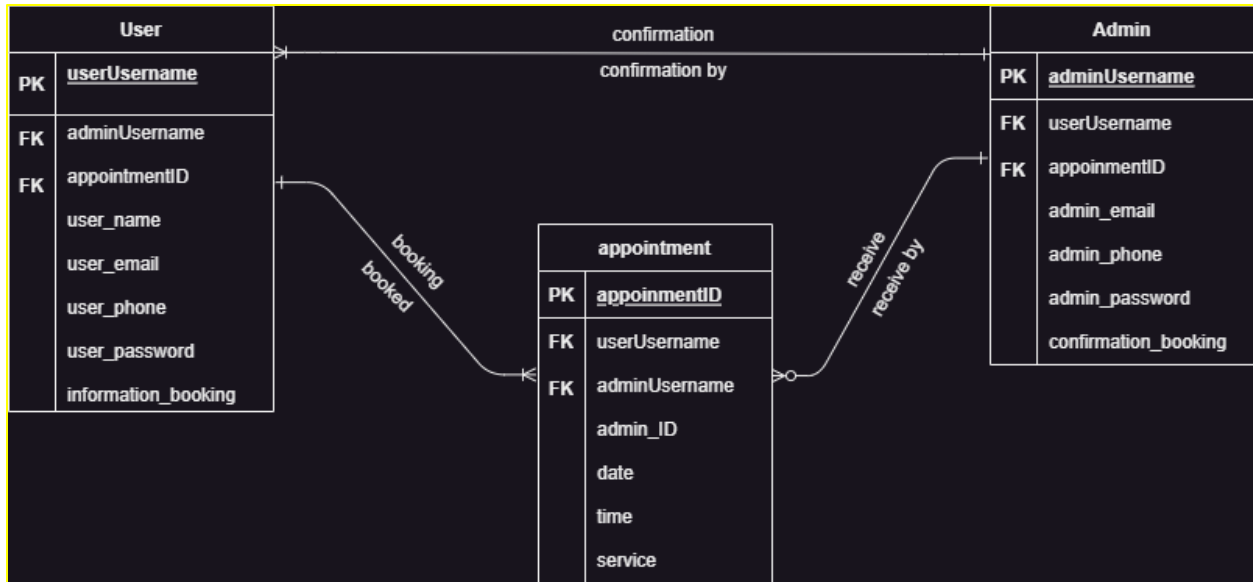


Figure 4.3.1(a): ERD Diagram

An Entity-Relationship Diagram (ERD) is a visual representation used in system analysis and design to illustrate the logical structure and relationships between entities (or objects) within a system. It helps to depict the entities, attributes, and the relationships between them in a clear and concise manner. In an ERD, there are four main components such as entity, attributes, relationships and cardinality.

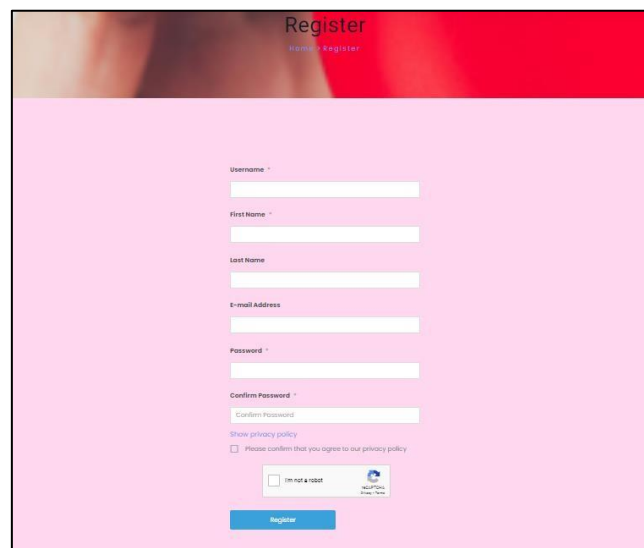
For our project ERD diagram has 3 entities. Entities represent the objects or concepts within the system being analyzed. They are typically represented as rectangles in the diagram. Examples of entities in our system are user, admin, and appointment. Each entity has a unique identifier, known as the primary key, which distinguishes it from other entities. Attributes, which are the properties or characteristics of an entity, are associated with each entity. For example, a user entity may have attributes such as user Username, name, email, and contact phone number. The user entity works as our patients who use the system and the admin entity is those who take care of the system.

The Cardinality and Modality of the user (customer) and admin account entity is, a user can sign up and login as a member of Booking Appointment Website. Next, the Cardinality and Modality of the admin and appointment entity is, an admin can give confirmation many bookings of Klinik Ceria. While the Cardinality and Modality of the user and appointment entity is, an user can book many types of appointments and service. The Cardinality and Modality of the Admin and user entity is, an Admin can manage the booking information of Klinik Ceria.

For the attributes in the ERD diagram, attributes are the specific properties or characteristics that describe an entity. They provide additional information about the entities in the system. Attributes can be represented within the entity rectangle or listed separately, depending on the style of the ERD. Each attribute has a name and a data type associated with it. For instance, in a user entity, attributes could include name, email, phone, password and information of booking details. Moreover, for the admin entity, including name, email, phone, password and confirmation of booking. Last for the attributes of appointment are ID, date, time, and service.

4.3.2 Input Design

4.3.2.1 User account management



The image shows a web registration form titled "Register" with a red header. The form is set against a light pink background and contains the following fields and elements:

- Username ***: A text input field.
- First Name ***: A text input field.
- Last Name**: A text input field.
- E-mail Address**: A text input field.
- Password ***: A text input field.
- Confirm Password ***: A text input field.
- Show privacy policy**: A link.
- ☐ Please confirm that you agree to our privacy policy
- ☐ I'm not a robot
- 
- Register**: A blue button at the bottom.

Figure 4.3.2.1.1

For those who do not already have an account, we provide a quick and user-friendly registration process, along with a captcha verification, to ensure secure access to Klinik CERIA's efficient appointment scheduling system. To become a registered member, simply enter your required information, including a unique username, your first and last name, a valid email address, and a strong password. Additionally, a captcha verification step is incorporated to confirm that you are a human user and not a bot, enhancing the security and reliability of the registration process.

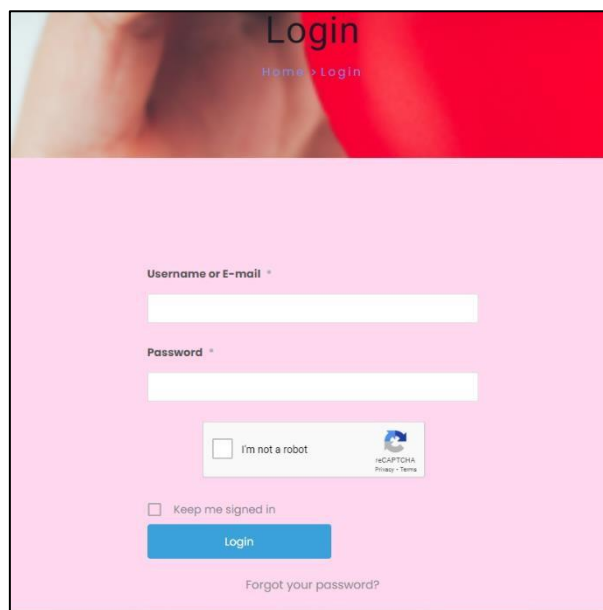


Figure 4.3.2.1.2

For registered members, simply log in by entering either your username or email address along with your password. Therefore, you be able to booking your appointment with Klinik CERIA through this system. As well as register section, Captcha verification step also incorporated as the security and reliability of the login process.

4.3.2.2 Booking system management

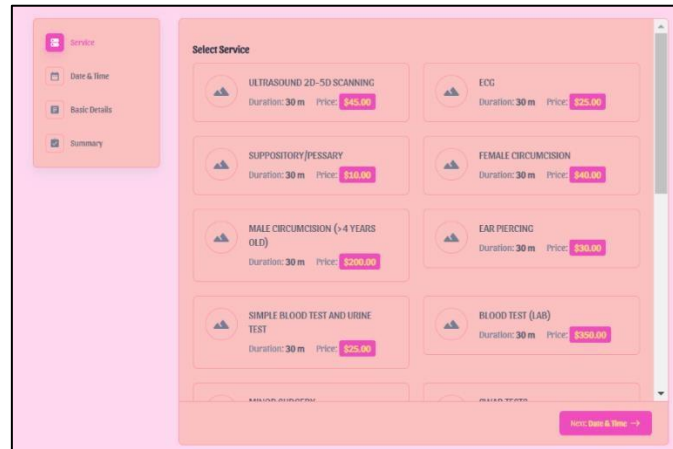


Figure 4.3.2.2.1

By choosing the precise service they want, patients may quickly schedule appointments. By offering a thorough list of services that are offered along with a pricing, patients can easily browse the possibilities and choose the one that best suits their healthcare needs. This simplified procedure guarantees that clients may properly and promptly schedule appointments for the particular services they want.

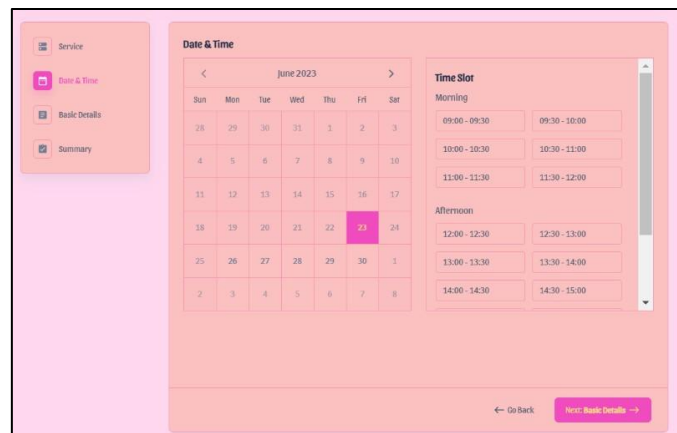


Figure 4.3.2.2.2

After selecting the desired service, patients can proceed to choose their preferred date and time for the appointment. Our intuitive appointment booking system provides a user-friendly interface where patients can easily navigate through a calendar to find available dates. Once a suitable date is selected, patients can then view the corresponding available time slots for that day.

The screenshot shows a web interface for a patient profile. On the left is a sidebar with four menu items: 'Service', 'Date & Time', 'Basic Details' (which is highlighted with a pink square), and 'Summary'. The main content area is titled 'Basic Details' and contains several input fields: 'Firstname' with the value 'Nurul Umami Shahlinah', 'Lastname' with 'Mehul Salfuddin', 'Email Address' with 'mehulasalfuddin2003@gmail.com', and 'Phone Number' with a dropdown for country code and the value '017-972 2854'. There is also a 'Note' field with the placeholder text 'Enter note details'. At the bottom right of the form, there are two buttons: 'Go Back' and 'Print Summary'.

Figure 4.3.2.2.3

Then, patients are required to provide their essential information in the patient profile section. This includes filling in their first name, last name, email address, and phone number. In addition to the basic profile information, patients also have the option to include any additional notes or specific requests they may have for Klinik Ceria. This allows patients to communicate their concerns or special requirements.

4.3.3 Output Design

4.3.3.1 User account management

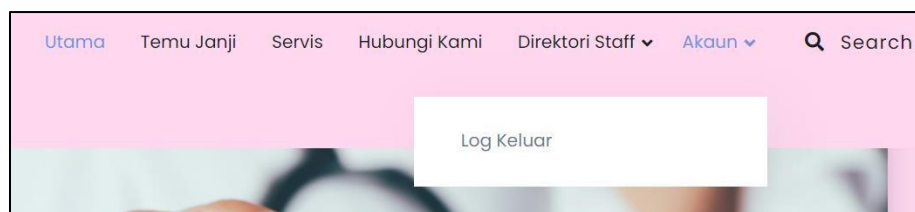


Figure 4.3.3.1.1

Once logged in to the website, you will notice a significant change in the account section. The previously displayed "Log/Register" option has now transformed into a "Log Out" button, indicating that you are successfully logged in and authenticated. With the transition to the "Log Out" button, you gain the privilege to utilize the appointment booking functionality effortlessly. You can easily navigate through the website and select your preferred services, dates, and time

4.3.3.2 Admin page management

[0] [Approved (6)] [Pending review (0)] [Waiting initial confirmation (0)] [inactive (0)] [Suspended (0)] [Admin tools]									
Administrator (6) Subscriber (6) BookingPress Customer (0) Clinic admin (1)									
Bulk actions Apply Change role to... Change Bulk Action Apply 0 items									
☐	Username	Name	Email	Role	Posts	Status			
☐	syndal	syndal mian	mushu@ny@gmail.com	Subscriber	0	Approved			
☐	balqasem	balqa.com	balqa@msb12@gmail.com	Subscriber, BookingPress Customer	0	Approved			
☐	bernard	peng peng	bern@gmail.com	Subscriber, BookingPress Customer	0	Approved			
☐	peng	0000 00	peng@gmail.com	Subscriber	0	Approved			
☐	york	carrie luo	york@gmail.com	Administrator	0	Approved			
☐	xavier	12 dtd	xavierxk@gmail.com	Administrator	0	Approved			
☐	xuan	yed e	xuanxu@gmail.com	Administrator	0	Approved			
☐	DRUMHUAJIA_87_47536	DRUMHUAJIA_87_47536 1234	xinhuo@qz@gmail.com	Clinic admin	0	Approved			
☐	bblooming@126	—	x1232323@chinauniversity	Administrator, BookingPress Customer	12	Approved			
☐	Username	Name	Email	Role	Posts	Status			
Bulk actions Apply Change role to... Change Bulk Action Apply 0 items									

Figure 4.3.3.2.1

As an administrator and clinic admin, they can access to view and manage the user database of the website. The admin has the ability to retrieve information about the users who have signed up or registered on the website. This includes details such as their usernames, email addresses, and other relevant information provided during the registration process.

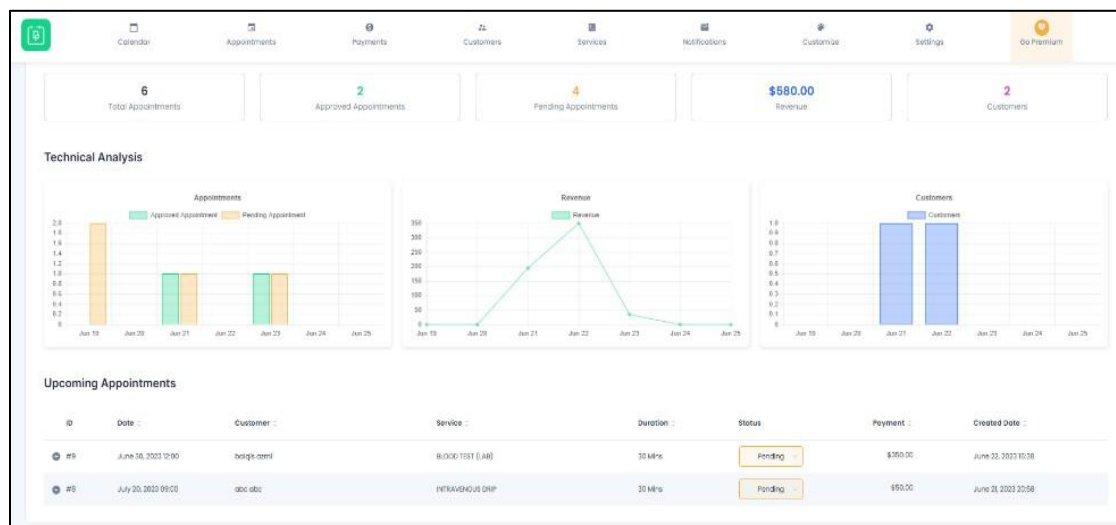


Figure 4.3.3.2.2

The administrator dashboard provides a comprehensive overview of the booking process, allowing the administrator to efficiently manage and track appointments. Within the dashboard, the administrator can access various sections, including the upcoming appointments, approved appointments, and the total number of appointments. In addition to managing the scheduling aspect, the dashboard also includes a feature that indicates whether a patient has made the required payment or not. This enables the administrator to easily identify if any outstanding payments need to be collected or if the payment process has been completed for each appointment.

4.3.3.3 Booking summary management

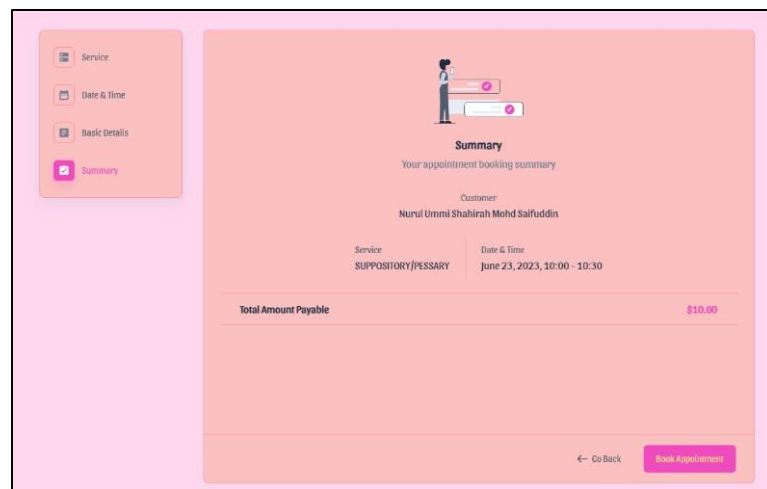


Figure 4.3.3.3.1

Before finalizing the appointment booking process, the website will display a comprehensive summary of all the details the patient has selected. This summary provides a clear overview, ensuring that patients can review and verify their choices before proceeding. Once satisfied with the selected service, date, time, and patient profile information, patients can simply click on the "Book Appointment" button located right below the summary.

4.3.3.4 Booking successful management

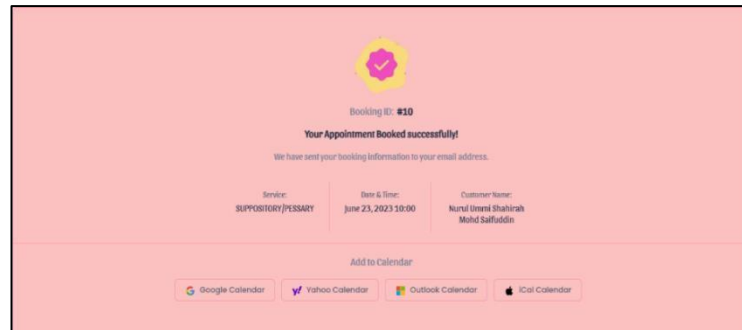


Figure 4.3.3.4.1

Upon successful completion of the booking process, the system will promptly display a confirmation message, notifying the patient that their appointment has been successfully booked. In addition to the confirmation message, our appointment booking system offers a convenient feature that allows patients to add a reminder to their preferred calendar application. Patients can effortlessly integrate their appointment details with popular calendar platforms such as Gmail, Yahoo, Outlook, and iCal.

Manage Appointments

+ Add New

Share URL

Need help?

Appointment Date

Customer Name

Service

Status

June 20, 2023

June 26, 2023

Start typing to fetch customer

Select service

Select status

Search for customers, services...

Reset

Apply

☐	ID	Date	Customer	Service	Duration	Status	Payment	Created Date
☐	#11	June 23, 2023 14:00	syahnuh roslan	ECG	30 Mins	Pending	\$25.00	June 23, 2023 10:45
☐	#10	June 23, 2023 10:00	Nurul Ummi Shahkiah Mohd Saifuddin	SUPPOSITORY/PESSARY	30 Mins	Approved	\$10.00	June 23, 2023 04:27
☐	#7	June 21, 2023 16:00	Nurul Ummi Shahkiah Mohd Saifuddin	HOUSE VISIT	1 Hours	Approved	\$100.00	June 21, 2023 15:13
☐	#6	June 21, 2023 15:30	yong yong	ULTRASOUND 2D-5D SCANNING	30 Mins	Pending	\$45.00	June 21, 2023 15:02

Figure 4.3.3.4.2

Once a patient successfully books an appointment, the system stores the relevant information in our databases. This ensures that all booked appointments are securely recorded and easily accessible. To ensure efficient management of appointments, the system provides administrative privileges to authorized personnel. The admin has the ability to access the booking details and

review pending appointments. The admin can then change the status of the patient's booking from "pending" to "approved" within the system.

4.3.4 Interface Design

4.3.4.1 Head Bar management



Figure 4.3.4.1.1

The head bar of our system features six distinct parts which each parts offering valuable information and functionalities for patients to explore. Patients have the flexibility to choose the specific section they want to access based on their preferences and needs.

4.3.4.2 Home page

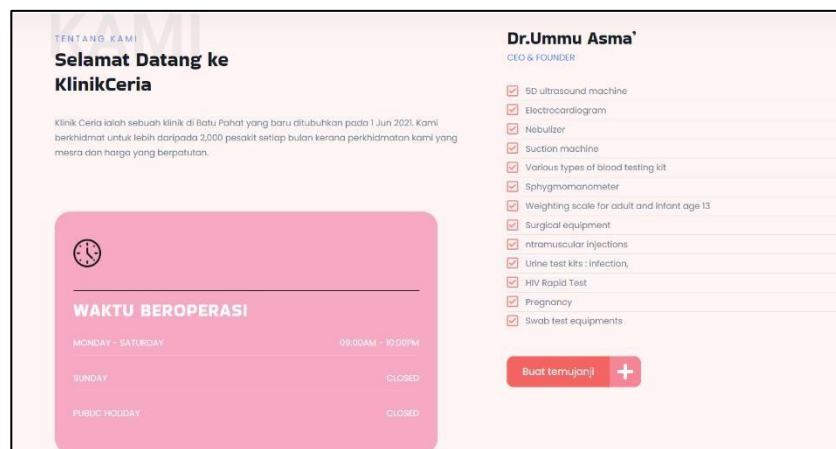


Figure 4.3.4.2.1

The home page of our system is designed to welcome patients with a warm introduction and provide them with essential information. Additionally, the home page prominently displays the clinic's working hours, ensuring that patients are aware of the specific times when Dr. Ummu and the clinic are available to provide medical care. Furthermore, a brief overview of the types of

services provided by Klinik CEO, Dr. Ummu, is showcased on the home page. By providing this overview, patients can quickly gauge the scope of healthcare services offered by the clinic.

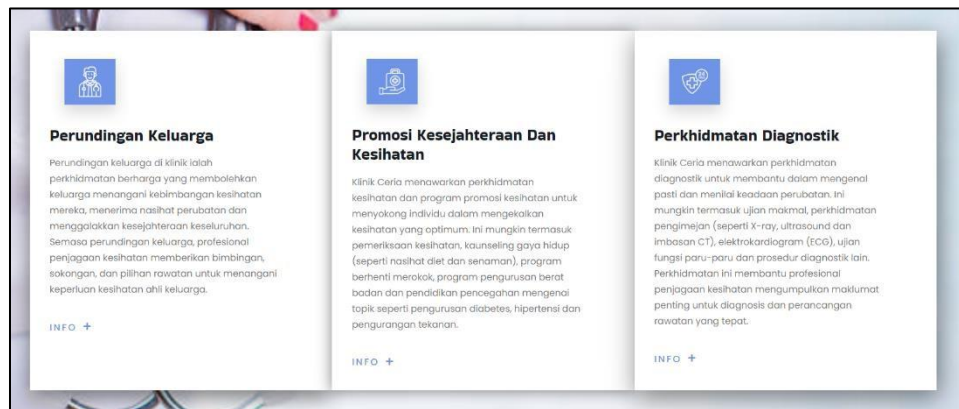


Figure 4.3.4.2.2

The homepage of Klinik Ceria also presents three compelling reasons why patients should choose our clinic as their primary healthcare provider. Firstly, we pride ourselves on being a family-friendly and children-friendly clinic. Secondly, we recognize the importance of convenience and accessibility. Lastly, Klinik Ceria provides a wide range of medical services, including various medication, investigations, and treatment options.



Figure 4.3.4.2.3

The homepage of Klinik Ceria features a prominent "Services" button, by offering patients a convenient way to explore the diverse range of medical services we provide. By clicking on the "Services" button, patients can gain comprehensive insights into the specific healthcare offerings available at our clinic. This dedicated section allows patients to delve deeper into the various services offered by Klinik Ceria for them.

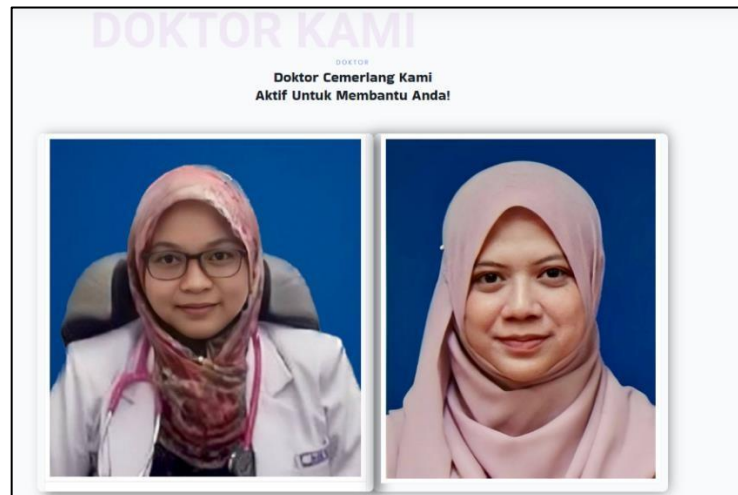


Figure 4.3.4.2.4

We also include picture of dedicated doctors who are committed to providing exceptional healthcare services. The primary doctor, Dr. Ummu, takes on a crucial role in overseeing and leading the clinic's medical operations. Additionally, Dr. Rozila as resident doctor at Klinik Ceria. will plays a vital role in delivering comprehensive healthcare services to our patients.

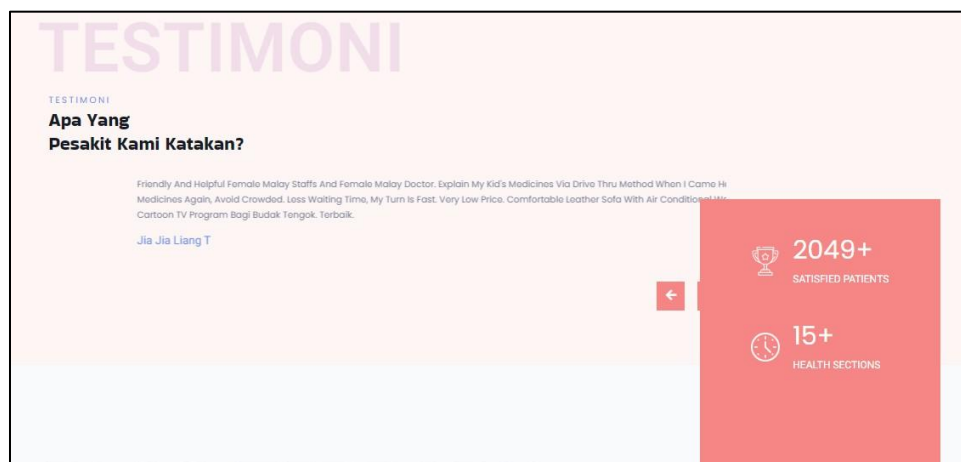


Figure 4.3.4.2.5

We understand the importance of transparency and providing insights into the patient experience at Klinik CERIA. That's why our homepage features testimonials from previous patients who have received care at this clinic. These testimonials allow prospective and new patients to gain a better understanding of how others perceive their experiences at Klinik CERIA.



Figure 4.3.4.2.6

At the end of our homepage, we provide the location details of Klinik CERIA to ensure ease of access for our patients. By including the clinic's address, we aim to make it convenient for patients to locate and reach us without any difficulties. Additionally, we have incorporated informative videos of Perkhidmatan Kesihatan Mesra Remaja (PKMR), serving as a valuable resource for patients. These videos are designed to provide patients with important information and guidance in case of emergency situations.

4.3.4.3 Booking appointment page

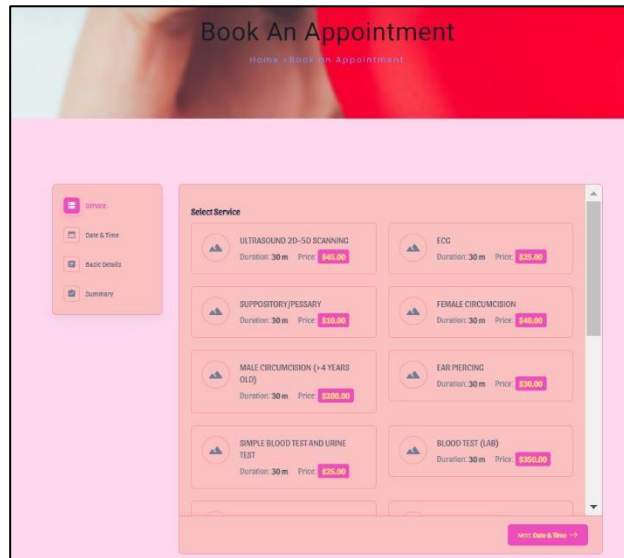


Figure 4.3.4.3

Here are the booking appointment page on the Klinik Ceria website which offers a user-friendly interface for patients to easily complete their appointment bookings. Patients can select their desired service, choose a convenient date and time, provide their patient profile details, review the appointment summary, and confirm the booking with a click. At last, the admin of website will send an notification which patient booking appointment is approved.

4.3.4.4 Services page

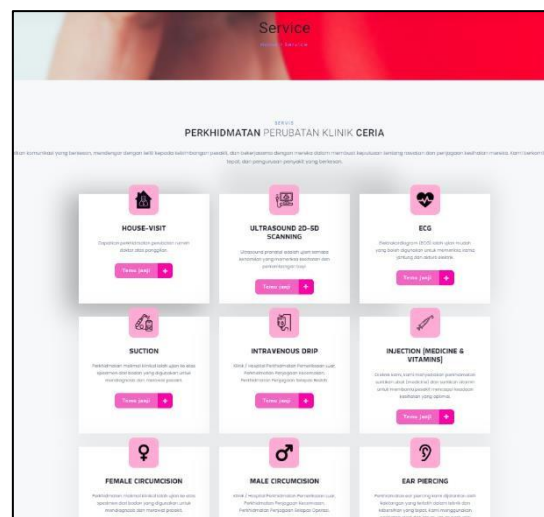


Figure 4.3.4.4.1

On the Services page of the Klinik Ceria website, patients can access detailed information about the range of services provided. Each service is presented with comprehensive details, enabling patients to make informed decisions about their healthcare needs. To enhance convenience, we have incorporated a "Booking" button for each service. Patients can simply click on the "Booking" button associated with their desired service, which will direct them straight to the booking appointment page. This seamless integration allows patients to easily transition from exploring the services to scheduling their appointments without any hassle.

4.3.4.5 Our team page

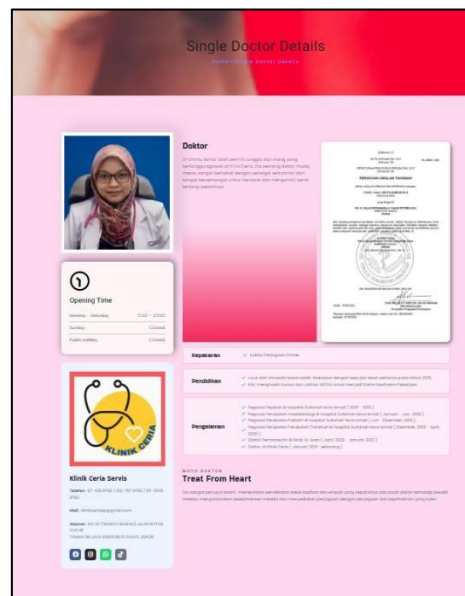


Figure 4.3.4.5.1

In the Team section of our website, we have two dedicated pages which are the Doctor page and the Staff page. On the Doctor page, patients can explore the profile of primary doctor, Dr. Ummu, and learn about her remarkable journey leading up to her role at Klinik Ceria. This section provides insights into Dr. Ummu's background and experiences, giving patients a deeper understanding of her expertise and qualifications as a healthcare professional.

media accounts. By simply clicking on these icons, patients are directed to the corresponding links or profiles. For example, clicking on the Facebook icon will take them directly to Klinik Ceria's official Facebook page, where they can access updates, announcements, and engage with the clinic's community. Similarly, the Instagram, TikTok, and WhatsApp icons will lead patients to Klinik Ceria's respective social media accounts or chat platforms, allowing for seamless communication and engagement.

4.3.4.7 Footer management



Figure 4.3.4.7.1

In the footer section, we have incorporated three essential components for easy navigation and information access. Firstly, we have the Useful Links section, which allows patients to directly access any page they desire on our website. Secondly, we provide the Working Days information, which informs patients about the operational hours of Klinik Ceria. This allows patients to plan their visits accordingly and ensures they are aware of the days when the clinic is open or closed. Lastly, we have the Reach Us section, which includes important contact details for Klinik Ceria. This includes the clinic's phone number, email address, physical location, and icons linking to our social media pages. Patients can easily reach out to us through the provided contact information or connect with us on our social media platforms, further facilitating communication and engagement.

4.4 Project Implementation

The implementation phase in the Software Development Life Cycle (SDLC) is where the Online Booking System for Klinik Ceria is developed, it must be built based on the design specifications. This is the most crucial phase that needs to be focused on because this phase will receive the most attention. The system is implemented at this phase to support the specific business need. This phase also will discuss part of the performance goals that are defined in the Planning phase and are used to measure the system's performance.

4.4.1 Requirement Specifications for Project Development

The Klinik Ceria online booking platform indicates simplifying appointment scheduling and improve operational effectiveness. Its goal is to provide a simple platform for people to easily schedule appointments. Doctors, staff, patients, and managers are the stakeholders for the system. The Requirement specification is the simple description of what the system must perform or the qualities it must have. There are two types of requirements in this project which are functional requirements and the non-functional requirement.

The functional and non-functional requirements that define the expected qualities and behaviors of the software system are fully defined in the required specifications for project development. These requirements provide a basis for the development phase and ensure the project's goals are fulfilled by the result that is achieved. Functional requirements is A functional requirement is a description of the service the system must provide. It indicates a software system or a part of it. A function is nothing more than the software system's inputs, behavior, and outputs. Non-functional requirements, often known as NFRs, are a group of specifications that outline the system's operational constraints and capabilities while also attempting to improve its functionality. These are effectively the requirements that define how well it will function, that involve things like performance, security, operational, cultural, and political aspects.

FUNCTIONAL REQUIREMENT	NON-FUNCTIONAL REQUIREMENT
• User account management - The system must be register before book the appointment - User must log in the online booking system before book the appointment - The system must allow user contact with staff at Klinik Ceria - The system should allow user to log out after use the system	• Operational - The online booking system will run on a platform that supports WordPress. - The online booking system should operate the online booking for Klinik Ceria - This system will have the database to store the schedule for an appointment - The system should operate consistently and reliable
• Online booking management - The system must display the available date for the patient to book the appointment - The system should have the reminder the patient about the appointment. - The system must provide a calendar for patients that are easy to book. - The system should allow the user to cancel and reschedule the appointment. - The system should be able to let user see the booking history	• Performance - The system can be available for 24 hours everyday. - The database can be store quickly after patient book the appointment - The interaction between patient and the system must be easy and fast.
• Service management - The system must have the list of service that Klinik Ceria provided - The system should allow user to select their service that they want - The system must have the service pricing information for each service. - The system must let the user give the review and ratings to the service that Klinik Ceria provides.	• Security - The patient information only the admin can see. - The system includes log in functionality for users. - The system user CAPTCHA to verify a user is a human or not.
• Staff management - The system must allow the staff to define the availability of doctor - The system must have staff profile including the doctor profile	• Cultural and political - The system uses Malay for people who are easy to understand.

Table 4.4.1. functional requirement and non-functional requirement

4.4.2 System Implementation Process

In order to transform the system design into an operational system, a booking appointment system must go through a number of processes in the system implementation process. The information system will be put into place by the analyst during this final stage of systems development. Training users on how to utilise the system is part of this step. Vendors provide some training, but the systems analyst is in charge of overseeing it. The analyst must also prepare for a seamless transition from the old system to the new one. This procedure entails converting files from outdated formats to new ones, creating a database, setting up the necessary hardware, and putting the new system into use. WordPress was the development tool we used for this project.

WordPress is a web content management system. It was originally created as a tool to publish blogs but has evolved to support publishing other web content, including more traditional websites, mailing lists, and Internet forums, media galleries, membership sites, learning management systems, and online stores. In this project, we chose to use WordPress as our system development tool as it is easier to operate and suitable for beginners like us who do not have any experience of developing a system. Next, by using WordPress, we can create a system online and anywhere. Besides, it also helps us to know the progress of our project. In addition, WordPress also has many functions to design the website, including the text, database and the functionality of each button. WordPress also provided a lot of templates for us to choose from so that we don't need to design the interface of a system on our own from scratch. This has made our job easier.

In conclusion, we have greatly shortened the time required to establish a booking appointment system for Klinik Ceria by choosing WordPress as our system development tool.

5.0 CONCLUSION

5.1 Achieving the Aims of the Programs

There are three objectives stated in this project. However, in this project we can only achieve two objectives. The objective that we can achieve if the Klinik Ceria is going to use the system is to solve the problem of Klinik Ceria having the Online Booking System and to help patients of Klinik Ceria facing the problem of manually booking.

Firstly, previously Klinik Ceria did not have an Online Booking System and only needed to book the appointment through WhatsApp and manual direct to Klinik Ceria. Now we have developed a new Online Booking System for Klinik Ceria and fulfill all the requirements that have been provided by Klinik Ceria. The system can work efficiently for patients to book the appointment. The patient also can look at the profile of the doctor and staff, the service Klinik Ceria provides and the pricing for each service. It also helps the staff to check the availability of the doctor that will be in charge on that day without manually checking through the traditional calendar.

Secondly, the objective that we can achieve if the Klinik Ceria uses our system is to help the patient facing problems to book appointment manually. The online booking system that we developed is user-friendly where the user or patient can easily book the appointment through the website without needing to go to the Klinik Ceria to book the appointment and easily reschedule or cancel the appointment if there is any problem. The patient also be able to see the date and time for booking the appointment.

Lastly, we cannot achieve the objective of increasing the performance of business since we have a lack of time and financial management. To achieve that objective is required to have a long-term strategy and result to achieve that objective.

5.2 Weaknesses of the system/project

Klinik Ceria never had an online booking system and a website so when developing a new system, there will be some weaknesses and flaws in the system. In our project, we did not manage to include an email notification when customers have confirmed their booking appointment. This will be an inconvenience for customers because they need to contact the clinic to double confirm their appointment. Next, there are no online payments for customers to be made so they have to walk-in to the clinic to make the payment after their appointment. Lastly, there is no language selection that customers can choose from on the website. We have asked the doctor which language they prefer for their website, and they prefer 3 languages which is Malay, English, and Chinese as most of their customers are Malays and Chinese. Hence, the website is displayed only in Malay so there will be few difficulties for customers that do not understand Malay.

5.3 Advantages of the system/ project

The system analysis and design process play a crucial role in the development of any project or system, providing a systematic approach to understanding, designing, and implementing solutions to meet specific business needs. The advantages of this process are numerous and can significantly impact the project's success.

Firstly, system analysis and design enable a comprehensive understanding of the problem at hand. Through careful analysis, the system analyst can identify the exact requirements, constraints, and objectives of the project. This helps in establishing a clear vision and scope for the system, ensuring that it aligns with the organizational goals and user needs. By gaining a deep understanding of the current system or process, potential bottlenecks, inefficiencies, and areas for improvement can be identified, leading to enhanced productivity and effectiveness.

Secondly, system analysis and design facilitate effective communication and collaboration among stakeholders. This process involves engaging with users, managers, subject matter experts, and technical teams to gather requirements, provide clarifications, and ensure that

everyone is on the same page. By promoting clear and consistent communication, it minimizes misunderstandings and conflicts, leading to a more efficient development process. Moreover, system analysis and design emphasize the importance of scalability and flexibility. By carefully examining the existing system and future needs, the analyst can design a solution that can adapt and grow with the organization. This ensures that the system remains relevant and valuable eventually, saving both time and resources by avoiding costly rework or replacements.

Furthermore, system analysis and design promote the identification and mitigation of risks. By conducting a thorough analysis, potential risks, such as technical challenges, budget constraints, or implementation issues, can be identified and addressed proactively. This helps in minimizing the chances of project failure or unexpected setbacks, increasing the overall success rate. Additionally, Klinik Ceria does not have a website for their company, and they use a manual system for booking the appointments. So, we hope by developing this system, it can advance their company in the booking section. Thus, it also can boost their business and people know what KlinikCeria has provided to the customers. System analysis and design foster a structured development process. It provides a roadmap for the entire project, outlining the various stages, milestones, and deliverables. This structured approach ensures that the project progresses in a logical and organized manner, reducing the likelihood of scope creep or missed deadlines. It also enables better resource allocation and tracking, enhancing project management and control.

In conclusion, system analysis and design offer numerous advantages in the development of projects or systems. By providing a comprehensive understanding, promoting effective communication, ensuring scalability, mitigating risks, and fostering a structured approach, this process significantly increases the chances of project success. It enables the creation of solutions that meet the needs of stakeholders, enhance productivity, and drive organizational growth.

5.4 Future Endeavour in the Project

In our project to develop the website for Klinik CERIA, we have focused on enhancing the user experience and streamlining the appointment booking process. Our future endeavors involve further improving the website and expanding our services. One potential initiative is the integration of an online chat or messaging feature, allowing patients to communicate directly with the clinic's staff in real-time for quick inquiries and assistance.

We also recognize the potential of telemedicine in transforming healthcare delivery. By expanding the functionality of the website, we aim to integrate telemedicine capabilities, which will allow patients to access virtual consultations and remote healthcare services. Telemedicine offers numerous benefits to both patients and healthcare providers. It provides convenience and flexibility by enabling patients to seek medical advice, receive diagnoses, and undergo follow-up appointments remotely. Through secure video calls, audio calls, or text-based messaging platforms, patients can connect with our healthcare professionals from the comfort of their own homes or any location with an internet connection.

Furthermore, enabling online access to test results will enable patients to review and understand their laboratory reports and diagnostic findings. They can track their progress, monitor any changes, and have a better understanding of their overall health status. This promotes a sense of ownership and involvement in their healthcare journey. By providing online access to medical records, patients can view their health information at any time, from anywhere. Having this information readily available empowers patients to actively participate in their own healthcare decisions and ensures accurate and up-to-date information is readily accessible to healthcare providers when needed.

To further enhance the convenience and accessibility of our services, we are considering the development of a dedicated application for Klinik CERIA. This application would serve as an additional platform for patients to access our services, schedule appointments, and stay updated with their healthcare needs. By developing an application, we aim to provide a seamless and user-friendly experience for our patient.

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Appendix

Interview form

Question	Answer
1)What type of language do you want to use on the website? English or in Bahasa Melayu?	Multilanguage but prefer Bahasa Melayu and Mandarin.
2) How many doctors are in the clinic?	2
3) Is there any specific doctor in the clinic such as the children's doctor?	No, but has female and male doctors
4)Could you provide a list of available time slots or doctor's schedules for each doctor?	As many slots as possible, one doctor is currently responsible for weekdays, and the other is responsible for the nights and weekends
5) How long are the operating hours? Does it open for up to 24 hours?	Usually, it takes 30 minutes for each patient. Sundays are the day off for Klinik Ceria.
6) What are the main services offered by Klinik Ceria?	We had almost all types of medication. For special medicine, we will try our best to find for customers at affordable prices
7) What type of website do you want us to design?	Minimal design that is not too crowded with information and it is easy to use for users.
8) How can patients access the website?	Patients can access by online platforms such as Facebook, google and WhatsApp
9) What database do you want to use for the website?	It should have name, ic number, gender, and phone number
10) Does Klinik Ceria want to add a calendar on the website?	Yes, it can be included in the website
11) What is the theme color of Klinik Ceria?	Light pink or pink colour.
12) Do you want that color to be the theme for the website?	Pink and white.

13) If a patient has booked an appointment, do you want to give the name of the doctor that charged on that day?	Yes, can give the name of the doctor in charge.
14) Who is the admin, Klinik Ceria, or the doctor?	Klinik Ceria is the admin.
15) Do you want to put any advertisements for Klinik Ceria on the website?	Yes, can put the advertisement for the Klinik Ceria website.
16) Do you want to add a robot chat on the website?	Yes, it can be put on the website
17) What kind of security features do you want in the system?	Just use simple features such as captcha verification
18) How many types of appointments can patients book on the website?	There is no limit to the types of appointments that patients can book
19) What kind of user feedback system do you want to implement on the website?	The patients can feedback through our clinic's email
20) What types of information do you need from patients when they book an appointment?	It should have the name, ic number, gender, and most importantly phone number